

Predicting Sprint Performance Using Race Progression Data

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Abstract

Sprint performance depends on the interaction of reaction time, acceleration, maximum velocity, and the ability to hold that velocity through the finish line. Final times are the standard measure of performance, but the measurements taken during a race carry additional information about how that time was produced. This research investigates the question: how accurately can machine learning models predict sprint performance using race progression data across the 100m, 200m, and 400m events? To answer this question, publicly available race records were preprocessed into three datasets containing reaction time, wind speed, and time and velocity splits for the 100m, 200m, and 400m races. Four supervised regression models—Linear Regression, Random Forest Regression, Gradient Boosting Regression, and HistGradientBoosting Regression—were trained to predict final times from early and intermediate race measurements. The datasets contained 2,631 samples for the 100m event, 2,345 samples for the 200m event, and 2,884 samples for the 400m event. Model performance was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 score. HistGradientBoosting Regression performed best on the 100m, Gradient Boosting Regression performed best on the 200m, and Random Forest Regression was the most accurate on the 400m. Taken together, these results indicate that race progression data carries substantial predictive information, and that the most effective model depends on the characteristics of the sprint event being analyzed.

1 Introduction

Sprinting events in track and field such as the 100-meter, 200-meter, and 400-meter races require athletes to accelerate hard, reach maximum velocity, and hold that speed through the finish line. Fractions of a second can make all the difference. Performance has traditionally been judged by an athlete's final time. However, a final time reveals nothing about how that result was produced, which means the information about which parts of the race mattered most is lost.

Race progression data, meaning the time and velocity splits recorded across a race, gives a fuller picture of an athlete's performance. For example, an athlete in the 100m who reaches maximum velocity early has a different sprinting profile from an athlete who accelerates more gradually but maintains speed more efficiently. If their final times are identical, the two athletes appear indistinguishable. Understanding the difference can help sprinters and coaches identify weaknesses and design more effective training regimens.

This project investigates the research question: how accurately can machine learning models predict sprint performance using race progression data across the 100m, 200m, and 400m events?

The goal of this research is to build predictive models capable of estimating final sprint times from early and intermediate race metrics. By examining which features contribute most to those predictions, this research also aims to show how different phases of a race influence its outcome.

The project is set up as a supervised machine learning regression task. The input data are the velocity and time splits for each race: every 10m for the 100m and every 50m for the 200m and 400m. The output of each model is a continuous numerical value, the athlete's final race time. Four regression algorithms were compared to determine which was most effective.

This research is personally meaningful to me because I run all three events for Saratoga High School. In my own races I am often behind in the first half and only gain ground in the second, which is what drew me to acceleration and race progression in the first place. Applying machine learning to the split data of the world's best sprinters combines my interest in track and field with data science.

2 Background

Previous research has shown that the phases of a race do not contribute equally to the final time, and that their relative importance varies by event. The 100-meter dash depends heavily on explosiveness out of the blocks and on reaching maximum velocity quickly, whereas the 200-meter and 400-meter place more weight on sustaining speed over a longer distance while managing fatigue.

Tønnessen, Haugen, and Shalfawi (2013) [1] examined reaction times among elite sprinters at world championships and found a significant relationship between reaction time and 100m performance. Their database covered 1,319 sprinters competing between 2003 and 2009. Although their work focused specifically on reaction time, it demonstrates the value of early-race measurements in explaining final performance.

Maćkała, Fostiak, and Kowalski (2015) [2] examined the relationships between kinematics, motor ability, anthropometric characteristics, and the initial (10m) and secondary (30m) acceleration phases of the 100m sprint. Their results indicate that the early phase of a sprint is closely tied to overall performance, though their sample was small (22 participants).

More recently, machine learning has been applied to sprint modeling. Tam and Yao (2024) [3] used Random Forest and Neural Network models to predict the velocity-time curve of the 100m, focusing on the acceleration phase, and evaluated them against a traditional exponential speed model. They also reported a strong negative correlation between maximum velocity and final 100m time. Their work shows that machine learning can capture nonlinear patterns in velocity data that are less visible in simple correlation analysis.

Previous studies have examined reaction time, velocity, and biomechanics, but have largely concentrated on the 100m. This project extends that work by applying the same modeling approach across the 100m, 200m, and 400m, in order to compare how race progression relates to final time in events of different lengths.

3 Dataset

Three datasets were assembled, one for each event. Each contains wind speed, reaction time, and split times with corresponding velocities at fixed distances appropriate to the event. Using the full progression rather than final times alone allows the models to observe how an athlete built, maintained, and managed speed across the race.

The initial dataset came from a publicly available GitHub repository containing sprint data already formatted for machine learning use. It was selected because it contained the required race variables in an organized form. To increase the sample size and the diversity of sprinters represented, the data was expanded using the elite race logs published by AthleteFirst [4].

Because AthleteFirst publishes its race databases as PDFs rather than CSVs, preprocessing was required. The race records were first extracted from the PDFs using MinerU, which converted them into markdown files. Those markdown files were then processed using Python code written with Gemini assistance to produce CSV files suitable for modeling.

The final datasets are summarized in Table 1.

Table 1: Dataset features across the 100m, 200m, and 400m events.

Event	# of Samples	# of Features	Feature Type
100 Meter	2631	26	Wind speed, reaction time, 10-meter split times and velocity measurements
200 Meter	2345	15	Wind speed, reaction time, 50-meter split times and velocity measurements
400 Meter	2884	22	Wind speed, reaction time, 50-meter split times and velocity measurements

The 100m dataset included measurements at 10m intervals in order to capture acceleration and maximum velocity within a short race. The 200m and 400m datasets used 50m intervals, which capture acceleration together with the speed maintenance required over longer distances.

The target variable was the athlete’s final time. The models therefore used the race progression measurements as inputs in order to learn the relationship between early and intermediate splits and final performance.

Each dataset was divided into training and testing subsets using an 80%/20% split, with `random_state = 42` so that the same split is reproduced on every run.

4 Exploratory Data Analysis

4.1 100m Event

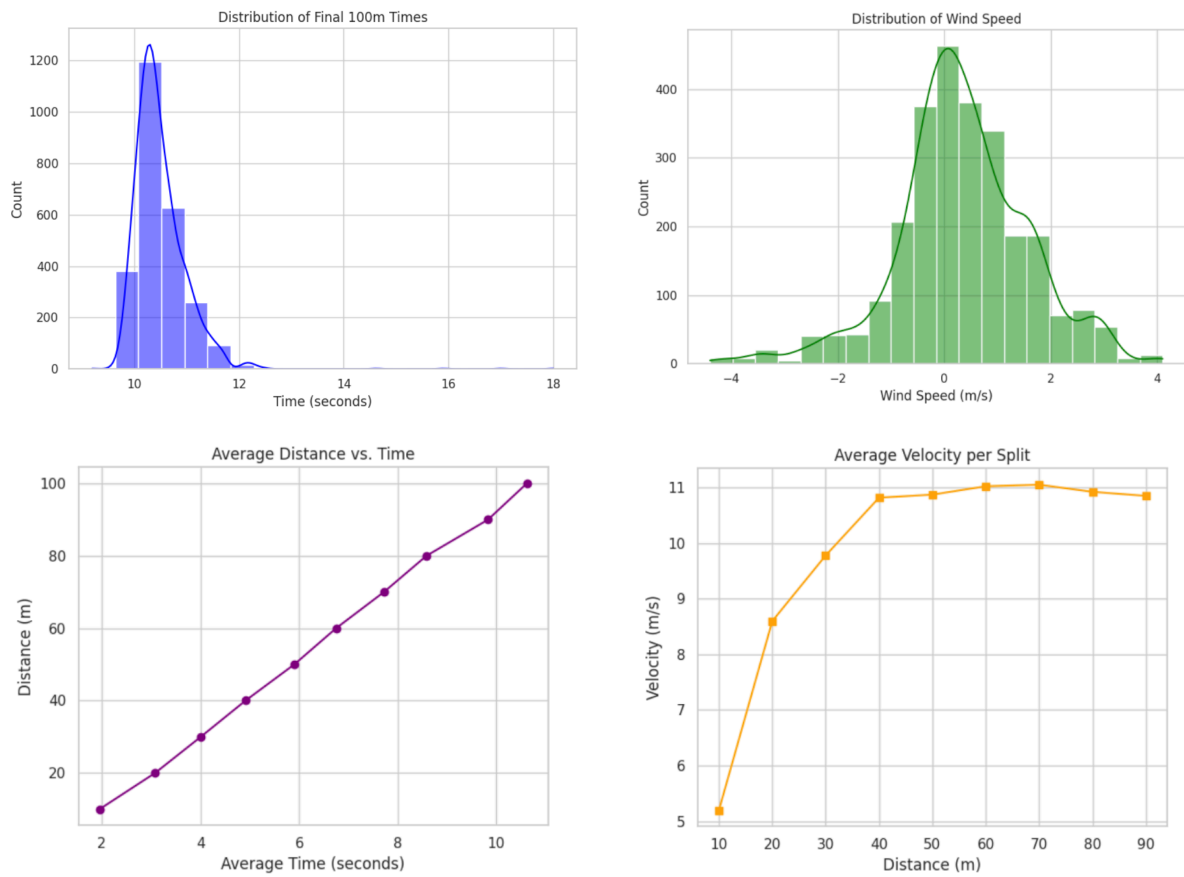


Figure 1: Data visualizations for the 100m event dataset.

4.2 200m Event

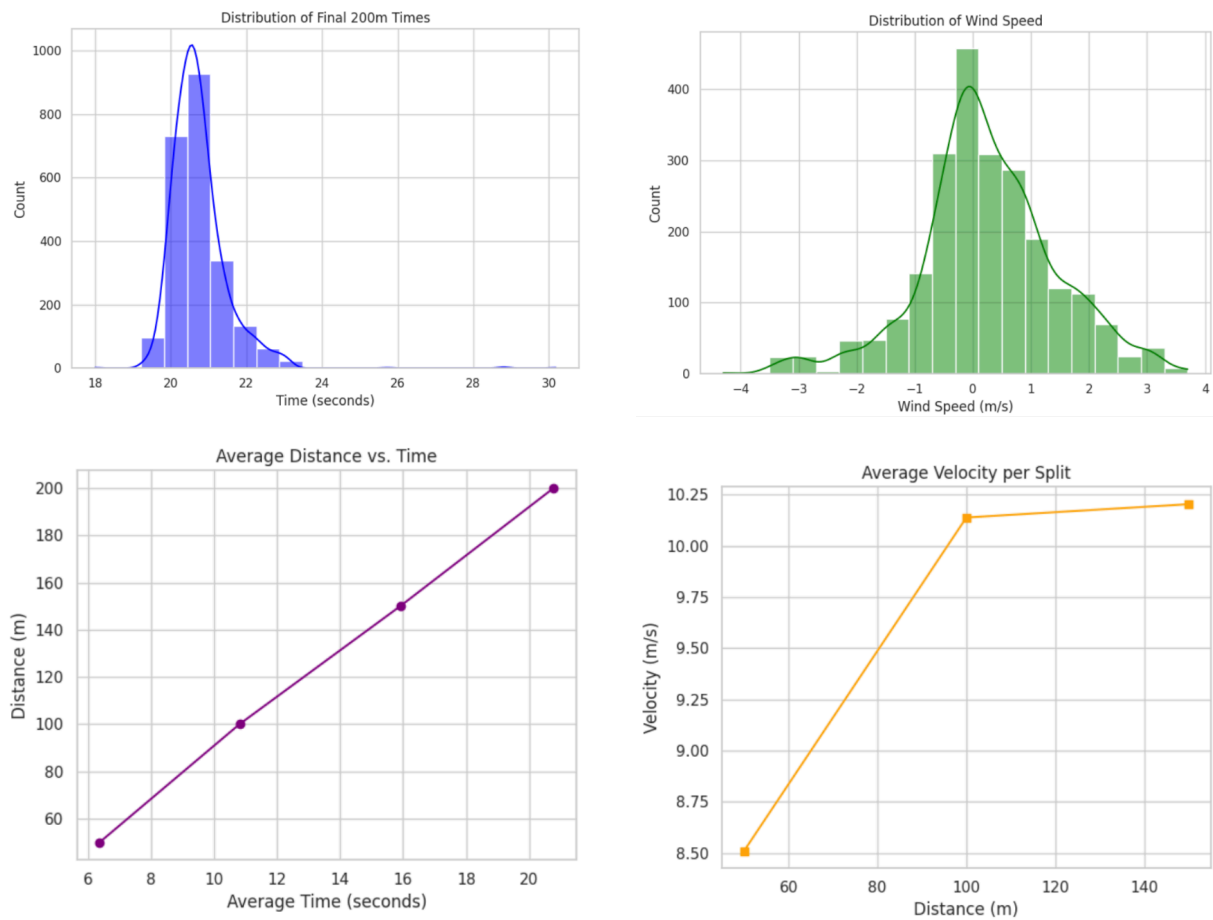


Figure 2: Data visualizations for the 200m event dataset.

4.3 400m Event

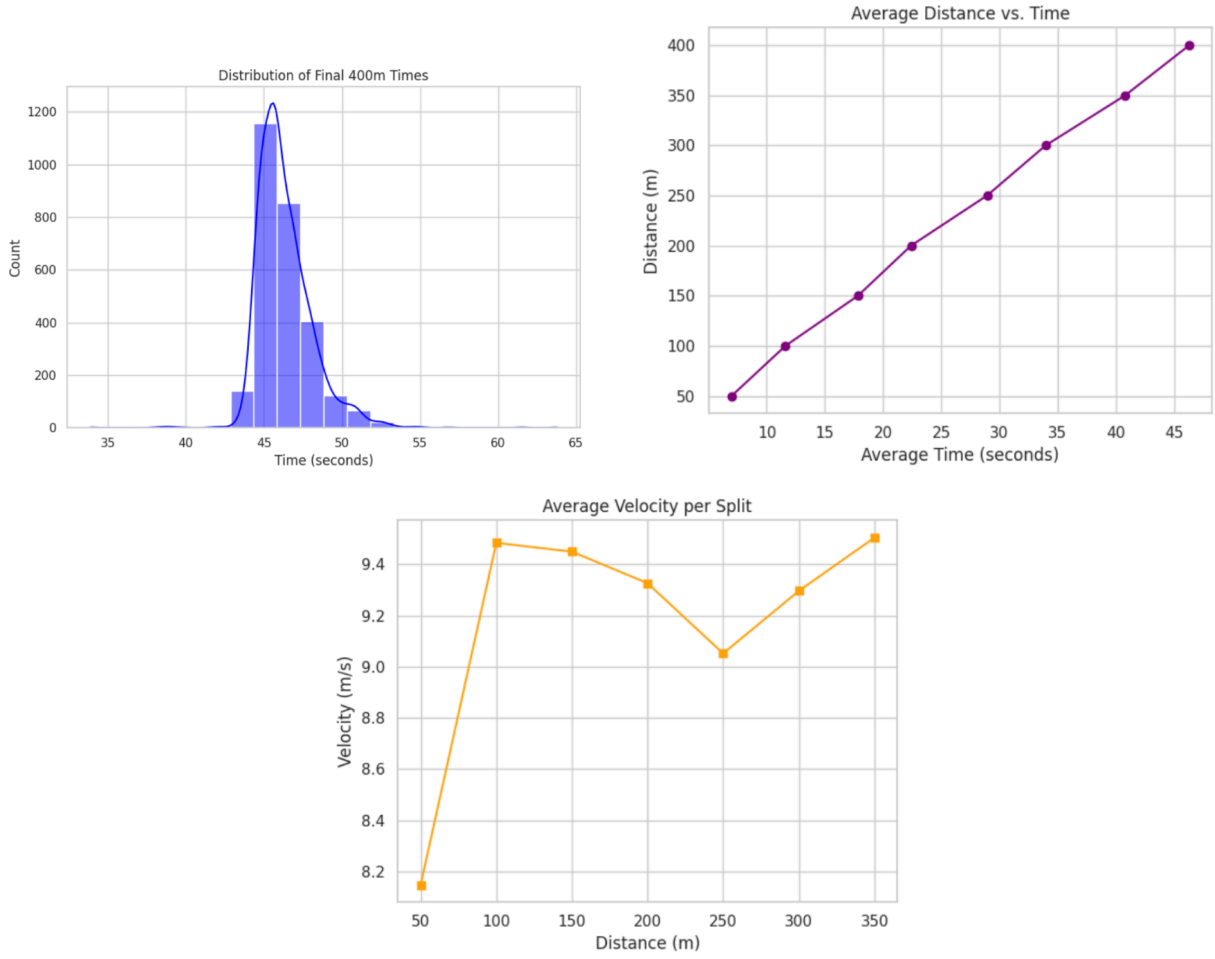


Figure 3: Data visualizations for the 400m event dataset.

5 Methodology / Models

This research used supervised machine learning to predict final times from race progression data. Because the output is a continuous measurement rather than a category, regression models were used. Four models were applied and compared: Linear Regression, Random Forest Regression, Gradient Boosting Regression, and HistGradientBoosting Regression.

Linear Regression was used as a baseline because it provides a simple model of the relationship between the race split inputs and the final time. It assumes this relationship can be expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \quad (1)$$

where y is the predicted race time, each x is a split time or velocity given to the model, and each β is a learned coefficient. Sprint performance is affected by nonlinear factors that Equation 1 cannot represent, but Linear Regression provides a useful benchmark for measuring how much additional accuracy the more complex models deliver.

Random Forest Regression combines the predictions of many decision trees. Each tree is trained on a bootstrap sample of the training data and considers a random subset of

features at each split, so the trees make partly independent errors. Averaging their predictions reduces overfitting and improves stability.

Gradient Boosting Regression builds a sequence of weak models, each fitted to the residual errors of the model before it. Each new model therefore focuses on the cases its predecessors handled worst.

HistGradientBoosting Regression was included for its efficiency on larger datasets. Instead of evaluating every unique value of a continuous feature when searching for split points, it first sorts feature values into a fixed number of discrete bins, which substantially reduces the number of candidate splits that must be tested.

The general workflow was:

1. Load and preprocess the sprint progression datasets.
2. Separate race progression features from the final times.
3. Split the data into 80% training and 20% testing subsets.
4. Train each regression model using the training data.
5. Generate predictions on unseen testing data.
6. Compare model performance using the evaluation metrics.

Model performance was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 score. All models were run with their scikit-learn default hyperparameters; no tuning was performed.

6 Results and Discussion

The aim of this research was to determine how accurately machine learning models can predict sprint performance from race progression data across the 100m, 200m, and 400m events.

Overall, the results indicate that machine learning models can predict final sprint times from race progression data. The strongest models achieved high R^2 scores, showing that velocity and time splits carry substantial information about the final result. No single model outperformed the others across all three events; the best-performing model varied with race distance.

100m Results: The 100m dataset produced the strongest predictive performance of the three events. HistGradientBoosting Regression performed best, with an MSE of 0.0108, RMSE of 0.1040, and R^2 of 0.9379. Full results are given in Table 2.

Table 2: Model performances for the 100m event.

Model	MSE	RMSE	R^2
Linear Regression	0.0530	0.2302	0.6958
Random Forest Regression	0.0157	0.1255	0.9096
Gradient Boosting Regression	0.0291	0.1706	0.8329
HistGradientBoosting Regression	0.0108	0.1040	0.9379

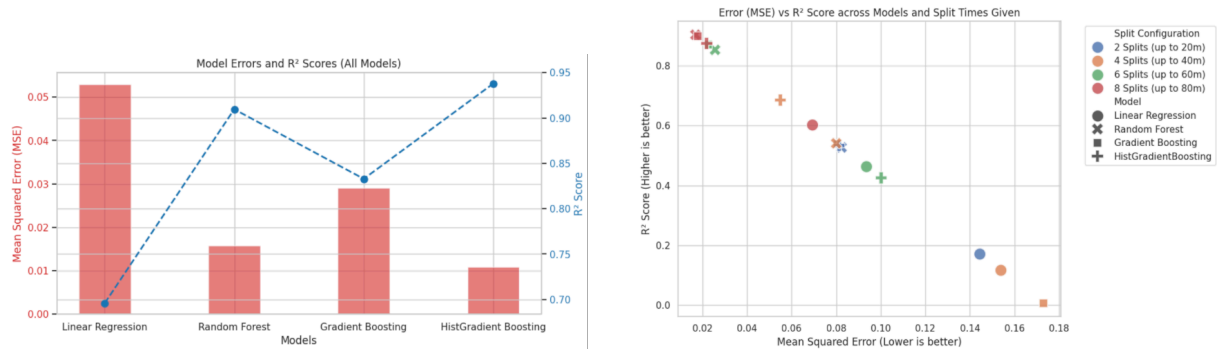


Figure 4: Prediction errors and R^2 scores across 100m models. The right panel shows how accuracy changes as fewer, earlier splits are supplied to each model.

200m Results: The 200m data produced lower predictive accuracy than the 100m and 400m datasets. Gradient Boosting Regression performed best, with an MSE of 0.1738, RMSE of 0.4170, and R^2 of 0.7946 (Table 3).

Table 3: Model performances for the 200m event.

Model	MSE	RMSE	R^2
Linear Regression	0.3018	0.5494	0.6435
Random Forest Regression	0.2325	0.4822	0.7253
Gradient Boosting Regression	0.1738	0.4170	0.7946
HistGradientBoosting Regression	0.2167	0.4655	0.7441

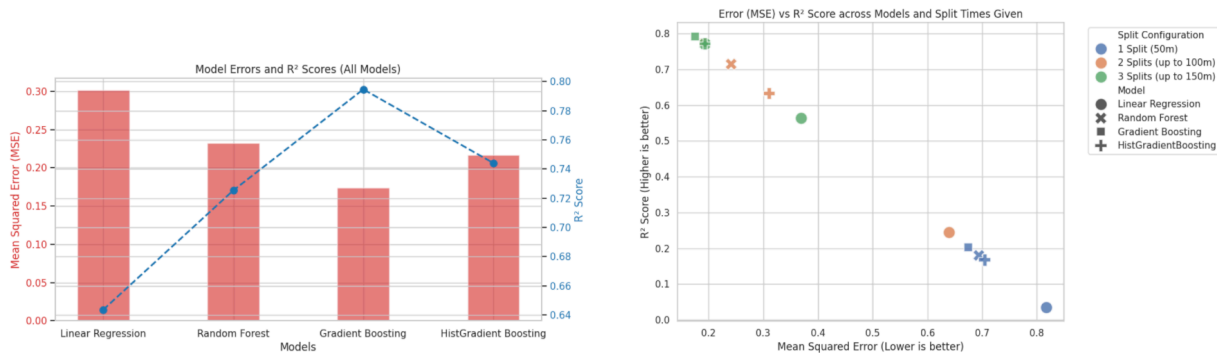


Figure 5: Prediction errors and R^2 scores across 200m models. The right panel shows how accuracy changes as fewer, earlier splits are supplied to each model.

400m Results: The 400m dataset also produced strong predictive performance, with Random Forest Regression the most accurate model (Table 4).

Table 4: Model performances for the 400m event.

Model	MSE	RMSE	R^2
Linear Regression	1.6546	1.2863	0.4502
Random Forest Regression	0.2469	0.4969	0.9180
Gradient Boosting Regression	0.3027	0.5502	0.8994
HistGradientBoosting Regression	0.3091	0.5560	0.8973

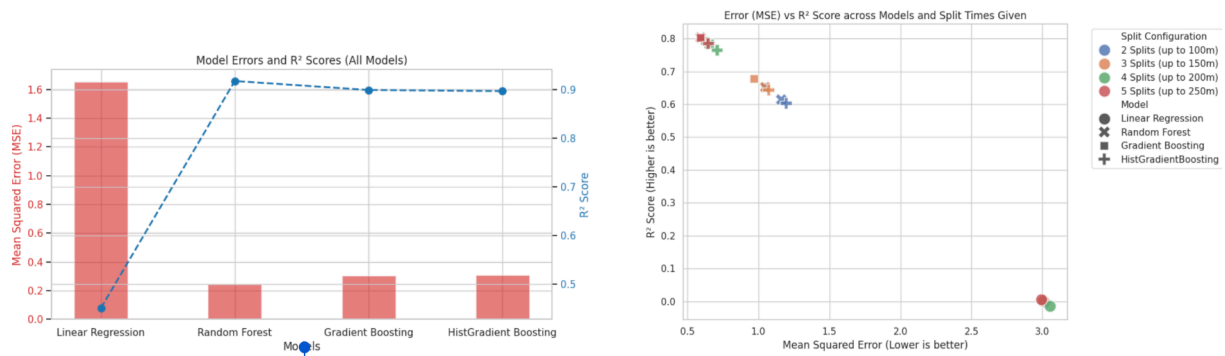


Figure 6: Prediction errors and R^2 scores across 400m models. The right panel shows how accuracy changes as fewer, earlier splits are supplied to each model.

Across the three events, the results show that different models suit different sprint distances. HistGradientBoosting Regression was most accurate on the 100m, where the race is dominated by acceleration and peak velocity. Gradient Boosting Regression was most accurate on the 200m, and Random Forest Regression on the 400m. One possible explanation for the pattern is that the longer events involve fatigue and pacing decisions that produce more complex interactions between race phases, though the present results do not test this directly.

The Linear Regression baseline performed noticeably worse on the 400m ($R^2 = 0.4502$) than on the other two events. Because linear models are sensitive to extreme values, and the 400m distribution contains times well above the main cluster, outliers are a plausible cause.

7 Limitations

Several limitations should be noted when interpreting these results.

First, the prediction task is partly constrained by arithmetic. When late splits are supplied as inputs, the final time is close to a deterministic function of those inputs, so high R^2 values do not by themselves demonstrate that the models learned meaningful structure. The split-configuration analysis in the right-hand panels of Figures 4, 5, and 6 is more informative in this respect, since it shows how accuracy degrades when only early splits are available.

Second, each model was evaluated on a single 80/20 train-test split with default hyperparameters. Without cross-validation, small differences between models cannot be distinguished from variation caused by the particular split.

Third, the datasets draw on elite competition records. Any conclusions therefore apply to elite sprinters and may not transfer to junior or recreational athletes, whose velocity profiles differ.

Fourth, the datasets pool male and female performances without modeling sex as a variable. Because velocity profiles differ systematically between these groups, part of the models' accuracy may come from implicitly distinguishing them rather than from race progression itself.

Finally, the models predict the end of a race from the middle of that same race. This is a different task from forecasting a future performance, which is what would be most useful to a coach.

8 Conclusion

This research examined whether machine learning models could accurately predict sprint performance from race progression data in the 100m, 200m, and 400m events. The results indicate that time and velocity splits carry sufficient information to predict final times with reasonable accuracy. HistGradientBoosting Regression was the most accurate model for the 100m, Gradient Boosting Regression for the 200m, and Random Forest Regression for the 400m.

These results suggest that different sprint distances favor different modeling approaches, which underlines the importance of accounting for race structure and event demands when constructing prediction models. Future work could address the limitations above by using cross-validation, separating athletes by sex, and comparing the models against a simple physical baseline such as extrapolating from the last observed velocity. Additional variables such as biomechanical measurements, height, and weight could also be incorporated.

References

- [1] Tønnessen, E., Haugen, T., & Shalfawi, S. A. I. (2013). Reaction Time Aspects of Elite Sprinters in Athletic World Championships. *Journal of Strength and Conditioning Research*, 27(4), 885–892. <https://doi.org/10.1519/JSC.0b013e31826520c3>
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- [3] Tam, C. K., & Yao, Z.-F. (2024). Advancing 100m Sprint Performance Prediction: A Machine Learning Approach to Velocity Curve Modeling and Performance Correlation. *PLOS ONE*, 19(5), e0303366. <https://doi.org/10.1371/journal.pone.0303366>
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